

3-Level Geometric Modeler: CSG, Subdivision Trees and Boundary Representation

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1 Definitions

1.1 Regularized Set

A set $S \subset \mathbb{R}^3$ is a regularized set $\iff S = \overline{\text{int}(S)}$.

1.2 Solid

Our goal is mainly focused on solids. A solid S is defined as a regularized subset of $\mathbb{R}^3$ ($S \subset \mathbb{R}^3$)

1.3 Polyhedron

A polyhedron P is the intersection of several half spaces. And the half spaces can be described as a set $H = \{h_1, h_2, h_3, \dots, h_n\}$

A polyhedron P contains features which can be described as three sets: a Vertex Set $V = \{v_1, v_2, \dots, v_m\}$ in which $v_i (1 \leq i \leq m)$ represents a vertex of the polyhedron P , $E = \{e_1, e_2, \dots, e_n\}$ in which $e_j (1 \leq j \leq n)$ represents an edge of P , $F = \{f_1, f_2, \dots, f_p\}$ in which $f_k (1 \leq k \leq p)$ represents a face of P .

2 CSG objects

Just as a polyhedron can be described as the intersection of half spaces, a CSG (Constructive Solid Geometry) model describes a polyhedron as Boolean operation on primitives which are the simplest solid objects used

for the representation. The Boolean operations on primitives are intersection, union, difference and complement. The primitives are solids. Suppose the equation of a surface is $f(x, y, z) = 0$, the primitive decided by this surface are either $f(x, y, z) \leq 0$ or $f(x, y, z) \geq 0$. For a polyhedron we define primitives as half spaces. But in more complicated cases, we need complex primitives like a cylinder, a sphere and so on.

2.1 Primitives:

These primitives comes from semigeometric definition and all these primitives are subsets of $\mathbb{R}^3$.

1. Half Space: A half space H is a subset of $\mathbb{R}^3$, H is defined by an inequality. Suppose the equation of a plane in $\mathbb{R}^3$ is $f(x, y, z) = a * x + b * y + c * z - d = 0$, which is a linear function of coordinates (x, y, z) , then the upper half space defined by equation $f(x, y, z) = 0$ are $f(x, y, z) = a * x + b * y + c * z - d \geq 0$, likewise the lower half space defined by equation $f(x, y, z) = 0$ are $f(x, y, z) = a * x + b * y + c * z - d \leq 0$.
2. Sphere : A sphere S should also be a subset of $\mathbb{R}^3$. The boundary of the sphere are defined by an equation $f(x, y, z) = (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 - r^2 = 0$. When $f(x, y, z) \geq 0$, we define the outer space of the

sphere including the boundary. Or when $f(x, y, z) \leq 0$, we define the inner half space including the boundary.

3. Cylinder : A Cylinder C are a subset of $\mathbb{R}^3$. The boundary of this cylinder are defined by an equation $f(x, y, z) = 0$

3 Subdivision trees:

1. Box: a Box is a solid. A box B in $\mathbb{R}^3$ is a subset of $\mathbb{R}^3$, $B = \{(x, y, z) | x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$, which is a regularized subset of $\mathbb{R}^3$

2. ? Box of a polyhedron P :

- (a) Empty Box: a box with no intersection with the polyhedron P , which means that all points inside the Box are exterior points of the polyhedron P .

A box B is an empty box $\iff B \cap P = \phi$.

- (b) Edge Box: An edge $e \in E$ bounds two faces $f_i, f_j \in F$, the edge and these two corresponding faces cannot disappear even if we subdivide the box into smaller ones. The endpoints of the edge lies on the boundary of the box. The intersection between the box and features of P other than e, f_i, f_j are an empty set.

So a box B is an edge box $\iff \exists e \in E, f_i, f_j \in F$, s.t. $B \cap e \neq \phi$, $B \cap f_i \neq \phi, B \cap f_j \neq \phi, \forall e_k \in$

$E, e_k \neq e, B \cap e_k = \phi, \forall f_l \in F, f_l \neq f_i, f_l \neq f_j, B \cap f_l = \phi$.

- (c) Vertex Box: A vertex v bounds edges and also bounds several faces $(f_1, f_2, \dots, f_n)$. And the vertex and the corresponding faces cannot disappear even if we subdivide the box into smaller ones. v is inside the box. The intersection between the box and features of P other than $v, f_1, f_2, \dots, f_n$ are an empty set.

So a box B is a vertex box $\iff$

- (d) Face Box: A face f bounds a polyhedron P . This box should contain exactly a face cutting through the box into two parts. The intersection of the face and the boundary of the box are either a triangle or a parallelogram.

A box B is a Face Box $\iff \exists f \in F$, s.t. $B \cap f \neq \phi, \forall f_i \in F$, if $f_i \neq f, B \cap f_i = \phi$.

- (e) Full Box: a box wholly inside the polyhedron, which means that all points inside the Box are interior points of the polyhedron.

A box B is a full box of Polyhedron $P \iff B \cap P = B$

3. Adequate subdivision tree: In $\mathbb{R}^3$, a subdivision tree are a octree in which each node of the tree would be a finite Box. The root are a finite box B_0 . We can construct the tree by beginning with the B_0 , and subdividing the tree until all leaves are ? Boxes.